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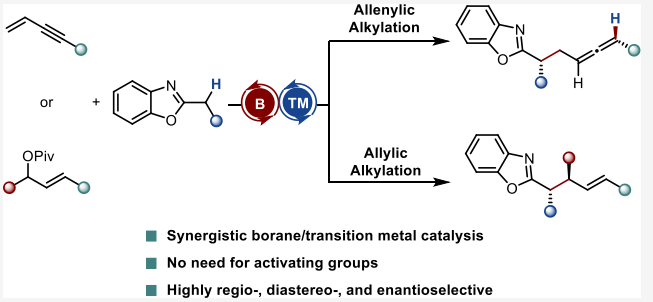
Borane/Transition Metal–Catalyzed Allenylic and Allylic Alkylation of Unactivated 2-Alkylbenzoxazoles

Qi-Fei Liu, Qian-Qian Peng, Sheng-Shi Hao, Lu Liu, Fu-Hao Zhang, Zhu-Jun Shi, Jian-Hua Xie, Chong-Ren Ai,* and Xiao-Chen Wang*

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ABSTRACT: Asymmetric functionalization of unactivated alkyl-substituted azaarenes at the α-position of the alkyl substituent is challenging because deprotonation of the weakly acidic α-C–H bonds requires a strong base. This requirement often impedes effective enantiocontrol by chiral catalysts. As a result, the few reactions that have been reported have narrow substrate scopes. Herein, we report a method for highly diastereo- and enantioselective allenylic alkylation reactions of unactivated 2-alkylbenzoxazoles with conjugated enynes, enabled by synergistic borane/nickel catalysis. The success of this method hinges on catalytic activation of the 2-alkylbenzoxazoles by a triarylborane, which permits deprotonation by triethylamine and thus is crucial for achieving both high yields and diastereo- and enantioselectivities. Extending the utility of synergistic borane/transition metal catalysis, we also demonstrate that a borane/palladium system effectively overcomes the limitations of palladium-catalyzed allylic alkylation reactions of unactivated azaarenes. In particular, previously unreactive acyclic 1,3-disubstituted allylic esters are rendered viable electrophiles, enabling the construction of products bearing two adjacent stereocenters.



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INTRODUCTION

Azaarene moieties substituted with alkyl groups containing carbon stereogenic centers are common in pharmaceuticals, agrochemicals, and natural products.[1] Direct functionalization of azaarenes at the α-position of a linear alkyl substituent via deprotonation with a base and electrophilic trapping of the resulting carbanion is the most straightforward strategy for constructing such structures. Although non-enantioselective variants of these reactions that afford racemic products are well developed,[2] catalytic enantioselective counterparts have largely been limited to azaarenes activated by electron-withdrawing groups[3−5] (Scheme 1a), which can be readily deprotonated with a relatively weak base (e.g., DBU or NEt_3). In contrast, deprotonation of unactivated azaarenes requires a strong base (e.g., ^{n}BuLi or lithium bis(trimethylsilyl)amide), use of which increases not only the likelihood of uncatalyzed background reactions but also the possibility of chiral catalyst decomposition. Notably, the few successful examples of catalytic asymmetric reactions involving unactivated azaarenes have primarily been metal-catalyzed allylic alkylation reactions, typically employing metal-π-allyl electrophiles generated in situ.[6] Other classes of electrophiles are rarely reported.[7] To overcome these limitations and expand the range of compatible electrophiles, alternative strategies are required.

Hydrometalation of conjugated enynes has been reported to generate *exo*-alkylidene-metal-π-allyl electrophiles that can be coupled enantioselectively with various nucleophiles to afford structurally diverse allenes in the presence of a chiral metal catalyst (Scheme 1b).[3c,8] However, reported examples of such coupling reactions are limited to relatively acidic pronucleophiles, which can be readily deprotonated by a relatively weak base to generate the corresponding nucleophile. The resulting protonated base may then shuttle the proton either to the metal catalyst, forming a metal hydride that subsequently hydrometallates the enyne,[3c,8a,b,d,e] or directly to the metalated enyne;[8c] both pathways ultimately afford the electrophile. In contrast, deprotonation of less acidic pronucleophiles requires stronger bases and generates proton-locked conjugate acids that are ineffective at transferring the proton. This inhibition of electrophile formation renders unactivated alkyl-substituted azaarenes unsuitable as pronucleophiles in metal-catalyzed reactions with conjugated enynes.

Based on these analyses, we hypothesized that strategies to enhance the α-C–H acidity of unactivated alkylazaarenes would enable their deprotonation by a relatively weak base,

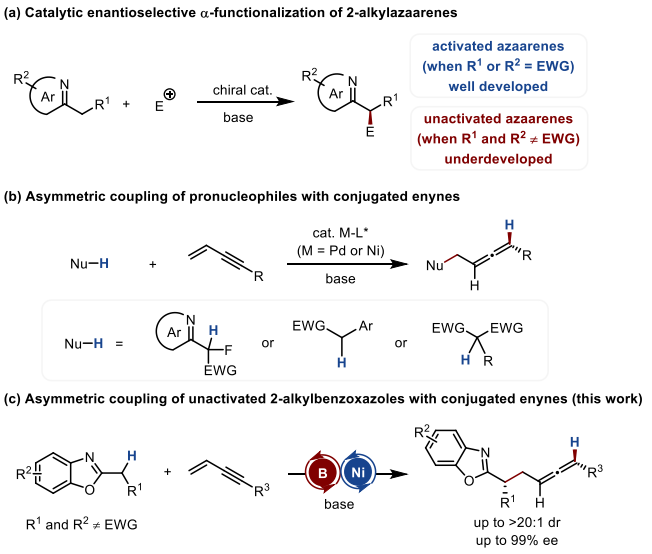
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Scheme 1. Catalytic Enantioselective α-Functionalization of 2-Alkylazaarenes

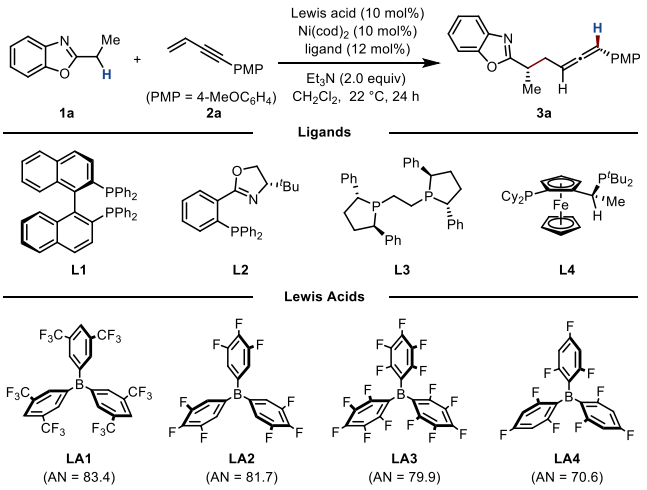


thereby allowing coupling with conjugated enynes under metal catalysis. Furthermore, employing a chiral metal catalyst could render the transformation enantioselective, thereby broadening the range of compatible electrophiles for the enantioselective α-functionalization of unactivated azaarenes. Notably, the Wasa group and our group recently developed various synthetic methods based on increasing the acidity of pronucleophiles by means of borane activation.[7f,9] In particular, in a prior study, our group demonstrated that a borane catalyst not only can activate azaarenes but also remains catalytically competent despite strong coordination by the nitrogen atom of the azaarene.[7f] This finding can be attributed to steric hindrance around the borane center, which promotes frustrated Lewis pair–type behavior.[10] We thus envisioned that borane activation of azaarenes could be utilized to accomplish unprecedented allenylic alkylation reactions of unactivated azaarenes with enynes. Indeed, we herein report that the desired transformation can be achieved by using a previously unreported synergistic borane/nickel catalytic system (Scheme 1c). The reactions exhibit a broad substrate scope and high diastereo- and enantioselectivities. In addition, we found that borane activation can be used for overcoming the limitations of palladium-catalyzed asymmetric allylic alkylation reactions of azaarenes.

RESULTS AND DISCUSSION

We commenced our study with the allenylic alkylation reaction of benzoxazole **1a** (1.0 equiv) with enyne **2a** (1.5 equiv) (Table 1). Initial screening focused on reactions involving chiral phosphine ligands **L1**–**L4** in the presence of triethylamine as the base and $B(C_6F_5)_3$ (**LA3**) as a Lewis acid catalyst (entries 1–4). (*R*)-BINAP (**L1**) and (*S*)-^{t}Bu-Phox (**L2**) showed no activity, whereas (*R*,*R*)-Ph-BPE (**L3**) and JosiPhos SL-J009-1 (**L4**) were catalytically active. Among them, **L4** provided the best outcome, affording product **3a** in 36% NMR yield with a 7:1 diastereomer ratio (dr) and 97% enantiomeric excess (ee) (entry 4). Subsequent screening of boron-based Lewis acids **LA1**–**LA4** (entries 5–7), which differ in Lewis acidity and steric bulk, in reactions with **L4** as the ligand revealed that the Lewis acid's properties markedly affected the

Table 1. Optimization of Reaction Conditions[a]



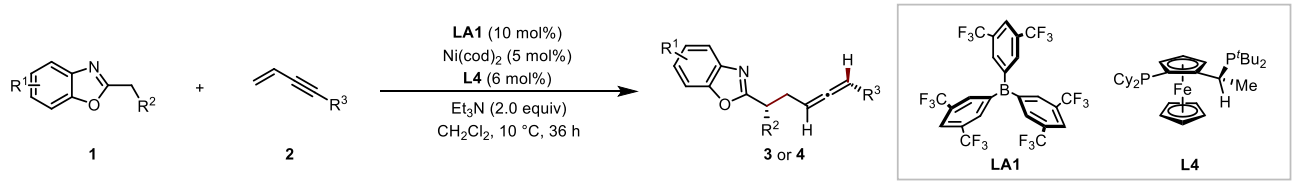
entry	ligand	Lewis acid	yield (%)	dr	ee (%)
1	**L1**	**LA3**	n.r.	n.a.	n.a.
2	**L2**	**LA3**	n.r.	n.a.	n.a.
3	**L3**	**LA3**	20	1:1	71
4	**L4**	**LA3**	36	7:1	97
5	**L4**	**LA1**	69	12:1	98
6	**L4**	**LA2**	23	4:1	96
7	**L4**	**LA4**	trace	n.a.	n.a.
8[b]	**L4**	**LA1**	87	>20:1	98
9[b,c]	**L4**	**LA1**	74	>20:1	98
10[b,c,d]	**L4**	**LA1**	85	>20:1	99

[a]Unless otherwise specified, all reactions were performed with 0.10 mmol of **1a**, 0.15 mmol of **2a**, 10 mol % $Ni(cod)_2$, 12 mol % ligand, 10 mol % Lewis acid, and 0.20 mmol of Et_3N in 0.20 mL of CH_2Cl_2 at 22 °C under N_2 for 24 h. Yields in entries 1–9 were based on ^{1}H NMR analysis; entry 10 reported the isolated yield. The dr and ee values were determined by HPLC using a chiral column. Abbreviations: n.r., no reaction; n.a., not applicable. [b]10 °C, 36 h. [c]5 mol % $Ni(cod)_2$, 6 mol % **L4**. [d]0.20 mmol of **1a**, 0.40 mmol of **2a**, 0.40 mmol Et_3N, 0.20 mL of CH_2Cl_2.

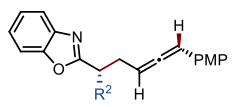
yield and diastereoselectivity. The use of **LA1**, which is sterically demanding and highly acidic (acceptor number (AN) = 83.4),[11] improved the yield from 36% (with **LA3**) to 69% and the dr from 7:1 to 12:1 (entry 5). **LA2**, which is less sterically demanding than **LA3** because of the absence of *ortho* fluorine atoms, gave a lower yield and poorer diastereoselectivity (entry 6). **LA4**, which is less acidic (AN = 70.6)[9c] than **LA3**, was inactive (entry 7). The stronger Lewis acidity of **LA1** may have resulted in better activation of **1a**, and this acid's steric bulk facilitated frustrated Lewis pair-type interactions with both **1a** and trimethylamine, interactions that were important for catalytic turnover. Lowering the reaction temperature from 22 to 10 °C and extending the reaction time from 24 to 36 h markedly enhanced both the yield and the diastereoselectivity and maintained the high ee (entry 8). Reducing the nickel catalyst loading from 10 to 5 mol % maintained the selectivity but modestly reduced the yield (entry 9). Notably, with 5 mol % of the nickel catalyst, scaling up the reaction from 0.1 to 0.2 mmol of **1a** and increasing the loading of **2a** to 2.0 equiv afforded **3a** in 85% isolated yield with >20:1 dr and 99% ee (entry 10). Control experiments confirmed that neither the borane catalyst nor the nickel catalyst alone produced the desired product (see Supporting

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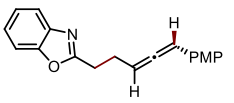
Table 2. Substrate Scope of Allenylic Alkylation Reactions between 2-Alkylbenzoxazoles and Enynes[a]



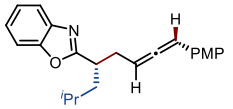
Scope of 2-Alkylbenzoxazoles



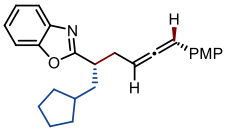
3a, R^2 = Me, 85% yield, >20:1 dr, 98% ee
3b, R^2 = Et, 84% yield, >20:1 dr, 98% ee[b]
3c, R^2 = ^{n}Bu, 82% yield, >20:1 dr, 98% ee[b]
3d, R^2 = Bn, 76% yield, 16:1 dr, 98% ee
3e, R^2 = ^{i}Pr, 20% yield, 1.5:1 dr, 97% ee and 93% ee[b]



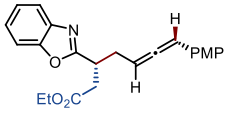
3f, 29% yield
92% ee[b]



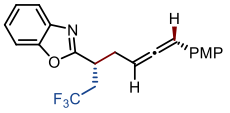
3g, 92% yield
>20:1 dr, 98% ee[b]



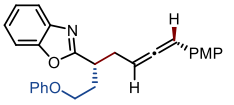
3h, 94% yield
16:1 dr, 98% ee[b]



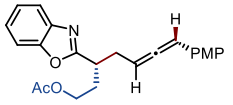
3i, 89% yield
14:1 dr, 98% ee[b]



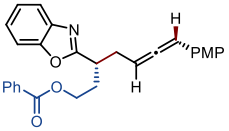
3j, 82% yield
14:1 dr, 98% ee[b]



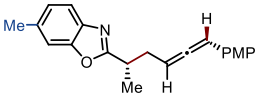
3k, 54% yield
20:1 dr, 98% ee



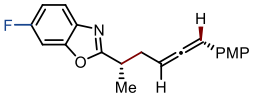
3l, 94% yield
19:1 dr, 98% ee[b]



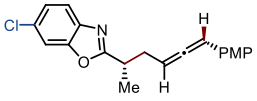
3m, 87% yield
>20:1 dr, 99% ee[b]



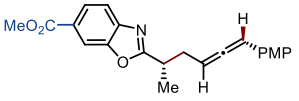
3n, 81% yield
15:1 dr, 98% ee[c]



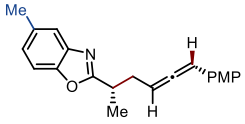
3o, 84% yield
19:1 dr, 98% ee



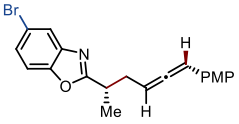
3p, 87% yield
>20:1 dr, 98% ee[b]



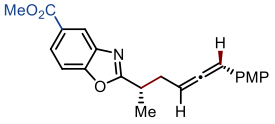
3q, 58% yield
14:1 dr, 98% ee[d]



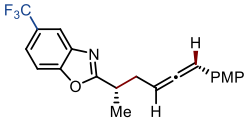
3r, 57% yield
>20:1 dr, 98% ee



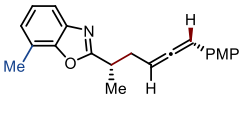
3s, 84% yield
>20:1 dr, 98% ee[d]



3t, 62% yield
>20:1 dr, 98% ee

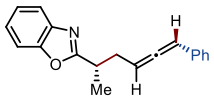


3u, 64% yield
20:1 dr, 98% ee

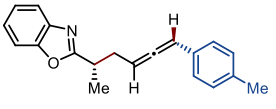


3v, 60% yield
19:1 dr, 98% ee[c]

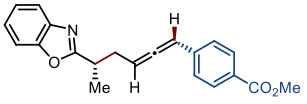
Scope of Enynes



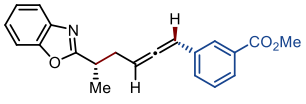
4b, 71% yield
>20:1 dr, 98% ee[d]



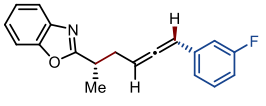
4c, 75% yield
>20:1 dr, 98% ee[d]



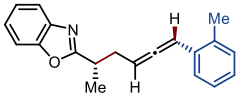
4d, 68% yield
19:1 dr, 98% ee[c]



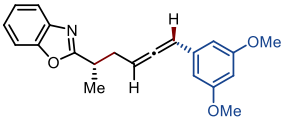
4e, 73% yield
14:1 dr, 99% ee[c]



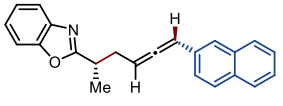
4f, 75% yield
13:1 dr, 98% ee[c]



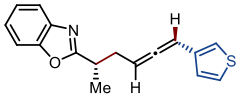
4g, 65% yield
10:1 dr, 98% ee[c]



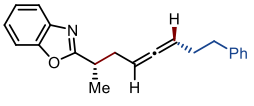
4h, 89% yield
16:1 dr, 99% ee[d]



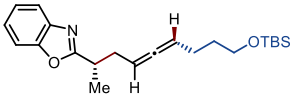
4i, 68% yield
19:1 dr, 98% ee[c]



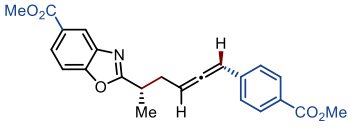
4j, 77% yield
14:1 dr, 97% ee[c]



4k, 61% yield
10:1 dr, 98% ee[c]



4l, 73% yield
11:1 dr, 99% ee[e]



4m, 65% yield
12:1 dr, 98% ee[d]

[a]Unless otherwise specified, all reactions were performed with 0.20 mmol of 2-alkylbenzoxazole, 0.40 mmol of conjugated enyne, 5 mol % $Ni(cod)_2$, 6 mol % **L4**, 10 mol % **LA1**, and 0.40 mmol of Et_3N, in 0.20 mL of CH_2Cl_2 at 10 °C under N_2 for 36 h. Isolated yields are provided. [b]10 mol % $Ni(cod)_2$, 12 mol % **L4**. [c]20 mol % **LA1**, 10 mol % $Ni(cod)_2$, 12 mol % **L4**, 22 °C. [d]10 mol % $Ni(cod)_2$, 12 mol % **L4**, 22 °C. [e]10 mol % $Ni(cod)_2$, 12 mol % **L4**, in 0.20 mL of $PhCF_3$.

Information), demonstrating that synergistic catalysis was essential.

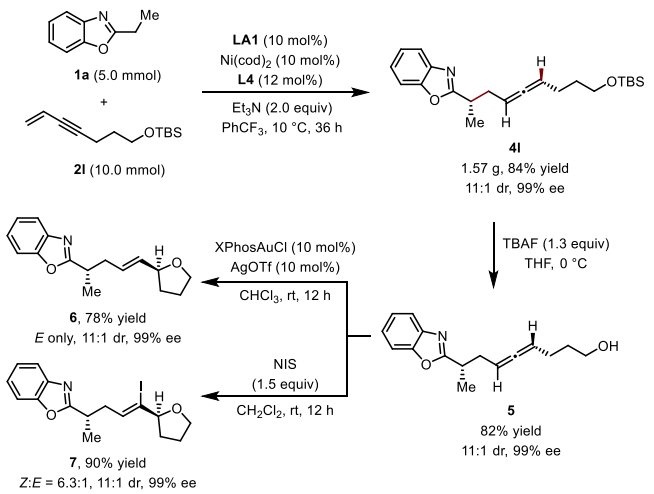
Having optimized the conditions for the allenylic alkylation reaction, we investigated its substrate scope (Table 2).

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Replacement of the 2-ethyl group in **1a** with ^{n}propyl, ^{n}pentyl, or phenethyl groups afforded products **3b**−**3d** in good yields with excellent diastereo- and enantioselectivities. When R^2 was ^{i}propyl, the increased steric bulk led to diminished reactivity and diastereoselectivity, affording product **3e** in 20% yield and 1.5:1 dr, while maintaining high ee values for both diastereomers. The use of 2-methylbenzoxazole also resulted in a low yield of the desired product (**3f**, 29%) along with a dialkylation byproduct in 10% yield (not shown in Table 2). In contrast, when the ^{i}propyl or cyclopentyl group was positioned one carbon further along the alkyl chain, the reactions proceeded smoothly to give **3g** and **3h** in high yields. Furthermore, 2-alkyl chains bearing terminal ester, trifluoromethyl, phenoxy, acetoxy, or benzoyloxy substituent were well tolerated, affording products **3i**−**3m** in 54−94% yields with excellent diastereo- and enantioselectivities (up to >20:1 dr and 99% ee). The diastereo- and enantioselectivity remained high across a range of benzoxazoles bearing diverse aryl ring substituents at various positions (**3n**−**3v**). Notably, benzoxazoles bearing ester substituents (**1q** and **1t**) failed to furnish the corresponding products (**3q** and **3t**) under conditions without **LA1**, indicating that the ester group could not provide the same activation as the borane catalyst. Aryl-substituted enynes bearing various electron-donating or electron-withdrawing groups at the *ortho*, *meta*, or *para* position of the benzene ring were compatible with the reaction conditions; products **4b**−**4h** were obtained in 65−89% yields with high diastereo- and enantioselectivities. 2-Naphthyl-, 3-thienyl-, and alkyl-substituted enynes also underwent the reaction, affording target products **4i**−**4l** with high diastereo- and enantioselectivities. The absolute configuration of **4m** was determined by X-ray crystallography of a single crystal. The scope of this reaction is presently confined to benzoxazoles (see the Supporting Information for incompatible heteroarenes).

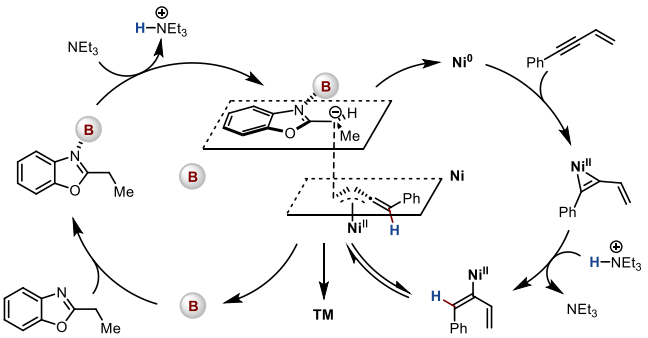
The allenylic alkylation reaction was readily scaled up: the reaction of **1a** (5.0 mmol) with **2l** (10.0 mmol) afforded **4l** in 84% yield with 11:1 dr and 99% ee (Scheme 2). Desilylation of **4l** with tetrabutylammonium fluoride provided **5** in 82% yield. Subsequent gold-catalyzed intramolecular cyclization[3c,12] of **5** via hydroxyl attack on the allene furnished **6** in 78% yield as the *E*-olefin exclusively, with retention of both dr and ee. In contrast, *N*-iodosuccinimide-mediated cyclization[13] of **5** gave **7** in 90% yield with the *Z*-olefin as the major isomer (Z/E = 6.3:1), also with retention of dr and ee.

Scheme 2. Gram-Scale and Subsequent Reactions



On the basis of literature precedents,[8c,9] we propose that these reactions proceed by a mechanism involving the interconnected catalytic cycles shown in Scheme 3. In the boron cycle, the borane catalyst enhances the acidity of the α-C−H bonds of the 2-alkylbenzoxazole by coordinating to the nitrogen atom. Subsequent deprotonation by triethylamine gives a borane-ligated carbanion (or aza-enolate),[14] which acts as a nucleophile in the coupling reaction. In the nickel cycle, the alkyne moiety undergoes oxidative addition to Ni(0) to generate a three-membered Ni(II)−olefin ring,[8c] which is subsequently protonated by the conjugate acid of the base to furnish a vinylnickel species. This species undergoes a rapid σ−π rearrangement to generate the key *exo*-alkylidene-Ni(II)-π-allyl electrophile. Outer-sphere attack by the nucleophile on the electrophile gives the product,[15] simultaneously regenerating the borane and nickel catalysts.

Scheme 3. Proposed Mechanism



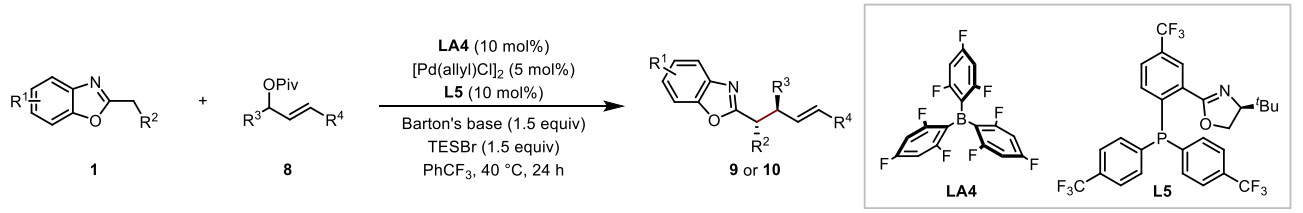
Inspired by the success of synergistic borane/nickel catalysis of allenylic alkylation reactions of unactivated 2-alkylbenzoxazoles, we explored the applicability of catalytic systems involving borane and other transition metals to asymmetric allylic alkylation reactions of unactivated azaarenes, which remain challenging, particularly for the construction of products with two adjacent stereogenic centers. To date, only two examples of the formation of compounds with vicinal stereocenters have been reported,[6a,b] both of which are restricted to cyclic allylic esters—likely because the steric hindrance associated with acyclic systems decreases their reactivity. We hypothesized we could overcome this challenge of poor reactivity of acyclic allylic esters by tuning the borane catalyst.

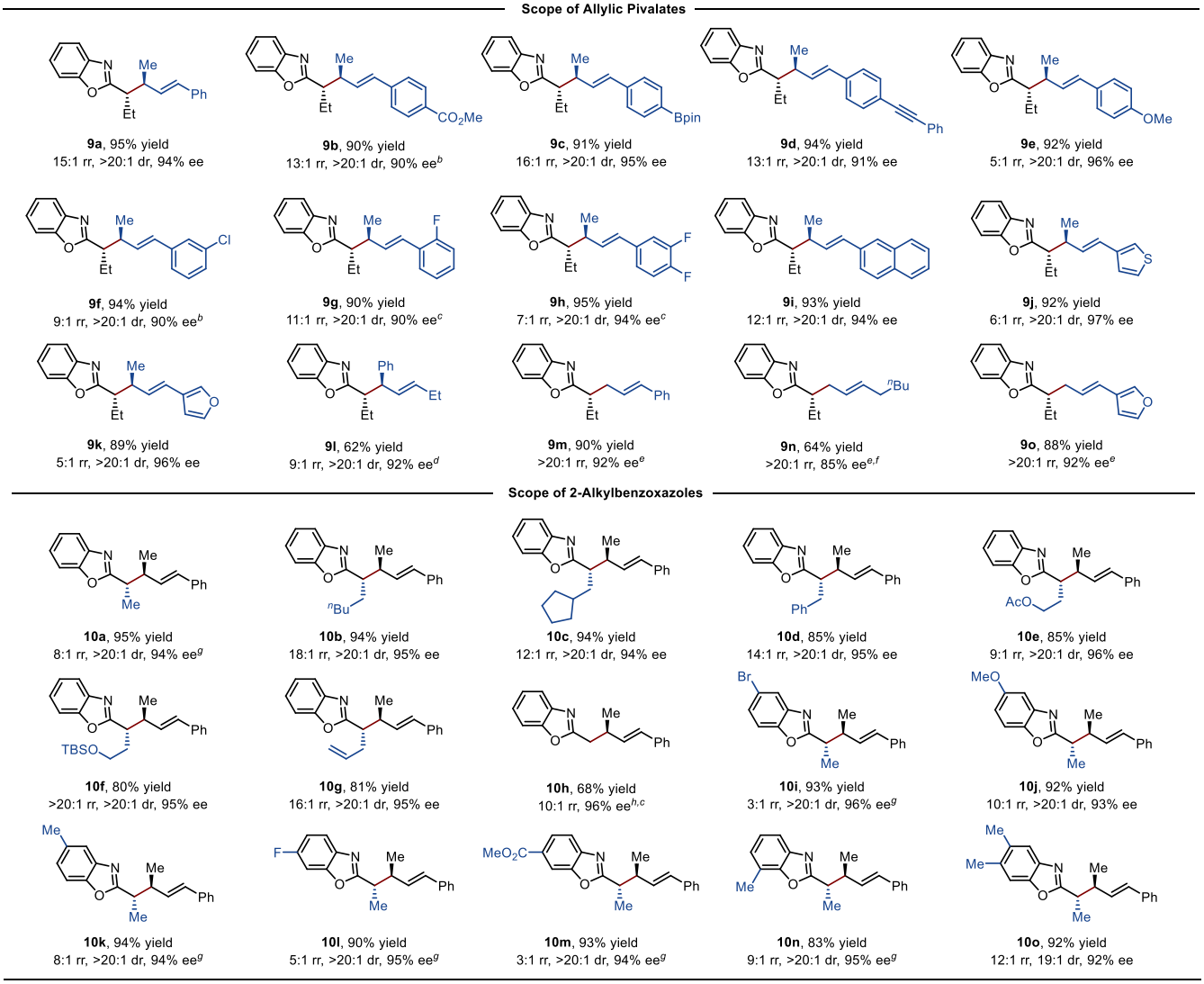
Through optimization of the reaction conditions (see Supporting Information), we found that a borane/palladium system efficiently catalyzed asymmetric allylation of 2-alkylbenzoxazoles with racemic 1,3-disubstituted acyclic allylic pivalates to form products with vicinal stereocenters (Table 3). The optimal conditions employed **LA4** (10 mol %), an in situ−generated Pd catalyst from $[Pd(allyl)Cl]_2$ (5 mol %) and **L5** (10 mol %), Barton's base (1.5 equiv), Triethylbromosilane (TESBr, 1.5 equiv) as an additive, and $PhCF_3$ as solvent, with the reaction carried out at 40 °C for 24 h under nitrogen.

Under these conditions, a wide range of racemic, acyclic 1-methyl-3-aryl-substituted allylic pivalates **8** bearing diverse functionality—including an ester; a pinacolboryl, alkynyl, or methoxy group; or one or more halogen atoms—reacted smoothly with 2-propylbenzoxazole, delivering products **9a**−**9h** in high yields with excellent diastereo- and enantioselectiv-

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Table 3. Substrate Scope of Allylation Reactions between 2-Alkylbenzoxazoles and Allylic Pivalates[a]





[a]Unless otherwise specified, all reactions were performed with 0.20 mmol of 2-alkylbenzoxazole **1**, 0.25 mmol of allylic pivalate **8**, 5 mol % $[Pd(allyl)Cl]_2$, 10 mol % **L5**, 10 mol % **LA4**, 0.30 mmol of Barton's base, and 0.30 mmol of TESBr in 1.0 mL of $PhCF_3$ at 40 °C under N_2 for 24 h. Isolated yields are provided. [b]Reaction temperature, 15 °C. [c]10 mol % **L10** (see Supporting Information). [d]10 mol % **L14** (see Supporting Information). [e]10 mol % **L2**. [f]0.40 mmol of allylic pivalate, 0.40 mmol of Barton's base, and 0.40 mmol of TESBr at −20 °C. [g]2.0 mL of $PhCF_3$. [h]0.40 mmol of allylic pivalate, 0.40 mmol of Barton's base, and 0.40 mmol of TESBr.

ities (≥20:1 dr, ≥90% ee) and 5:1−16:1 regioisomer ratios (rr). The absolute configuration of **9a** was determined by X-ray crystallography of a single crystal. The minor regioisomer was formed via addition of the nucleophile to the benzylic position of a Pd(II)-π-allyl species. Moreover, the aryl group of **8** could be a 2-naphthyl (**9i**), 3-thienyl (**9j**), or 3-furyl (**9k**) group without markedly affecting the yield or selectivities. Interestingly, when a 1-ethyl-3-phenyl-substitued allylic pivalate was used, addition at the benzylic position predominated, giving **9l** as the main regioisomer. Additionally, monosubstituted allylic pivalates gave only linear products (**9m−9o**) with one stereocenter.

Reactions of 1-methyl-3-phenyl-substituted allylic pivalate (**8a**) with 2-alkylbenzoxazoles bearing an ethyl or ^{n}hexyl group or a functionalized linear alkyl group afforded corresponding products **10a−10g** in 80−95% yields with high regioselectivity (8:1 to >20:1 rr), diastereoselectivity (>20:1 dr), and enantioselectivity (≥94% ee). Notably, when the 2-alkyl substituent was a methyl group, **10h** was obtained in 68% yield with 10:1 rr and 96% ee under modified reaction

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conditions. Furthermore, substrates with an electron-donating or electron-withdrawing substituent on the benzene ring of the benzoxazole were tolerated, giving corresponding products **10i**−**10o**.

Notably, the use of TESBr was crucial to the success of these allylic alkylation reactions; in the absence of this additive, the yield of **9a** was only 5% in otherwise identical conditions. Furthermore, using 1.0 equiv of **LA4** without TESBr resulted in a 60% yield of **9a** (see Supporting Information). Gas chromatography−mass spectrometry revealed that the standard reaction conditions afforded a considerable amount of TESOPiv. Because the pivalate anion coordinates more strongly than bromide to the borane catalyst, owing to the better interaction of the oxygen 2p lone pair with the empty boron 2p orbital, TESBr may enhance catalyst turnover by scavenging the pivalate anion after product formation.

CONCLUSIONS

In conclusion, we used synergistic borane/transition metal catalysis to overcome the challenges posed by asymmetric functionalization of unactivated alkylazaarenes at the α-position of the alkyl substituent. This strategy allowed us to achieve the first examples of coupling reactions between unactivated 2-alkylbenzoxazoles and conjugated enynes, which were promoted by a borane/nickel catalytic system and afforded previously elusive compounds with 1,3-nonadjacent stereocenters. Moreover, diastereo- and enantioselective asymmetric allylic alkylation reactions of unactivated 2-alkylbenzoxazoles with acyclic 1,3-disubstituted allylic esters were achieved with a borane/palladium system. A key element of both transformations is the use of fluorinated triarylboranes, which enable base-mediated deprotonation of otherwise poorly acidic α-C−H bonds under mild conditions. The method has a broad substrate scope, allowing reactions of diverse 2-alkylbenzoxazoles and electrophiles (>60 examples). Our work not only provides powerful solutions for accessing complex chiral architectures bearing multiple stereocenters from readily available precursors but also establishes synergistic borane/transition metal catalysis as a versatile strategy for asymmetric functionalization reactions.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.5c13835.

Experimental details, compound characterization data, HPLC chromatograms, DFT calculations, and NMR spectra (PDF)

Accession Codes

CCDC 2409963 and 2479536 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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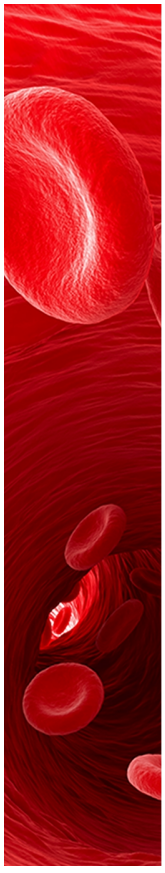
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(15) The schematic illustration of the relative positions and orientations of the nucleophile and electrophile in this addition step was deduced from the absolute configuration of the product and further corroborated by density functional theory (DFT) calculations. The calculations indicate that the steric bulk of ligand **L4** and the borane catalyst plays a dominant role in controlling the diastereo- and enantioselectivity.



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